



VADODARA INSTITUTE OF ENGINEERING

Lesson Planning – Theory

Subject: Advanced Java
Discipline: CE
Academic Year: 2025-2026
Employee code: 25CH0587

Subject Code: DI04016051
Semester: 4th
Name of Faculty: Prof. Naman Shah
Total no of lecture planned: 42

Teaching and Examination Scheme:

Teaching Scheme			Credits	Examination Marks					Total Marks
L	T	P	C	Theory Marks		Practical Marks			
				ESE (E)	PA (M)	ESE (V)		PA (I)	
					PA	ESE	OEP		
3	0	2	4	70	20	20	10	20	150

L- Lectures, T-Tutorial, P-Practical; C-Credit; ESE- End Semester Exam; PA- Progressive Assessment

Lecture	Monday	Tuesday	Wednesday	Thursday	Friday	Saturday
Turn on:	√	√			√	√

Lect. No.	Topic Covered	Book No.	Page No.	CT1	CT2	MSE/ RMSE	Planned Date	Actual Date	HOD's Sign	Remarks
1	Overview of Mobile Operating Systems: types and evolution	1	1	√		√	03/01			
2	Android and iOS comparison: features, architecture, and market share	1	6	√		√	05/01			
3	Features and Versions of Android: from early versions to latest	1	7	√		√	06/01			
4	Introduction to Android Studio: installation and IDE overview	1	9	√		√	07/01			
5	Running Apps on Emulator and Physical Device: setup and configuration	1	15	√		√	10/01			
6	Android Architecture Overview: all 5 layers introduction	1	16	√		√	16/01			
7	Linux Kernel Layer: role, features, and drivers	1	18	√		√	17/01			
8	Libraries Layer: SQLite, WebKit, OpenGL and others	1	19	√		√	19/01			
9	Android Runtime: Dalvik vs ART, DEX files	1	21	√		√	20/01			
10	Application Framework Layer: managers and APIs	1	32,33	√		√	21/01			
11	Applications Layer: built-in and third-party apps	1	35,36	√		√	24/01			
12	Application Components Overview: Activity, Service, Broadcast Receiver, Content Provider	1	37.38	√		√	27/01			
13	Activity and its Lifecycle: onCreate, onStart, onResume, onPause, onStop, onDestroy	1	59	√		√	28/01			
14	Service and its Lifecycle: started vs bound services	1	66	√		√	31/01			
15	Broadcast Receiver: concept, registration, and usage	1	67		√	√	02/02			
16	Content Provider: concept, URI, and CRUD overview	1	72		√	√	03/02			
17	Android Manifest: structure, permissions, and component declaration	1	83		√	√	04/02			
18	Revision & Practice: Architecture and Components unit	1	118		√	√	09/02			



19	Layouts: Linear Layout (horizontal and vertical orientation)	1	119		√	√	10/02			
20	Layouts: Relative Layout (positioning child views)	1	140		√	√	16/02			
21	Layouts: Constraint Layout (constraints and chains)	1	151		√	√	17/02			
22	Layouts: Grid Layout (rows, columns, and spans)	1	158		√	√	18/02			
23	Views & Widgets: TextView and EditText	1	154		√	√	21/02			
24	Views & Widgets: Button and ImageView	1	175		√	√	23/02			
25	Views & Widgets: ScrollView and ListView	1	159		√	√	24/02			
26	Views & Widgets: RecyclerView with Adapter and ViewHolder	1	179		√	√	25/02			
27	Concept of Event-Driven Programming in Android	1	186		√	√	28/02			
28	Event Listeners and Callback Methods	1	180		√	√	02/03			
29	Event Handling: onClick(), onKeyDown(), onTouch(), onFocusChange()	1	213		√	√	03/03			
30	Item Selection Events using onItemSelected() with ListView	1	246		√	√	04/03			
31	Intents: Explicit Intents and passing data between activities	1	247		√	√	07/03			
32	Intents: Implicit Intents and Intent Filters	1	251		√	√	09/03			
33	Toasts and Dialogs: AlertDialog, DatePickerDialog	1	256		√	√	10/03			
34	Menus: Options Menu, Context Menu, and Popup Menu	1	264		√	√	11/03			
35	Revision & Practice: UI Design and Event Handling unit	1	271		√	√	14/03			
36	Shared Preferences: storing and retrieving key-value data	1	282		√	√	16/03			
37	SQLite Database: SQLiteOpenHelper Class and database creation	1	290		√	√	17/03			
38	SQLite CRUD Operations: Insert and Read SQLite CRUD Operations: Update and Delete Firebase Realtime Database: setup and connection with Android Studio	1	290		√	√	18/03			
39	Firebase: Insert, Retrieve, and Display Data Revision & Practice: Data Storage unit Introduction to Kotlin for Android: syntax and comparison with Java Overview of Cross-Platform Development: Flutter and React Native	1	314		√	√	23/03			
40	Introduction to APIs and JSON Parsing; Overview of Publishing an Android App	1	314		√	√	24/03			
41	Revision	1	350		√	√	25/03			
42	Revision	1	350		√	√	01/04			
43	Revision	-	-		√	√	04/04			
44	Revision	-	-		√	√	06/04			



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Reference Books:

1. Advanced Java, M. T. Savaliya, dreamtech
2. Black Book “ Java server programming” J2EE, 1st ed., Dream Tech Publishers, 2008. 3. Kathy walrath
3. Complete Reference J2EE by James Keogh mcgraw publication

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